

Faculty Panel CSV — Column Glossary

Applies to: - `tenure_pipeline/R1_tenure_data.csv` - `tenure_pipeline/faculty_panel_advisor.csv` - Source: `tenure_pipeline/faculty_panel_with_pools.jsonl`

Generated from `panel_builder.py`, `pool_metrics.py`, and `openalex_resolver.py`.

Section 1 — Person Identifiers

Column	Type	Description
<code>name_display</code>	string	Full name as it appeared on the department faculty page (e.g. <code>"Jane A. Smith"</code>). Scraped from HTML; may vary slightly across years for the same person.
<code>name_key</code>	string	Normalized lowercase version of the name used for de-duplication and matching (e.g. <code>"smith_jane_a"</code>). Not intended for display.
<code>faculty_id</code>	string	Internal unique identifier assigned by the pipeline (slug + name hash). Stable within a pipeline run; used to link rows for the same person across years.
<code>openalex_id</code>	string	OpenAlex author identifier (e.g. <code>"A2345678901"</code>). Used to pull publication counts from the OpenAlex open scholarly index. Blank if no match was found (<code>match_confidence = NONE</code>).

Section 2 — Institution and Year

Column	Type	Description
<code>university</code>	string	Full institution name (e.g. <code>"Carnegie Mellon University"</code>).
<code>uni_slug</code>	string	Short machine-readable identifier for the institution (e.g. <code>"cmu"</code>). Used for joins and file naming.
<code>year</code>	integer	Calendar year of the observation. Each row represents one person–year. Data window is approximately 2000–2023.

Section 3 — Academic Rank

Column	Type	Description
rank	string	Rank of the person in that year, as parsed from the faculty page.

Rank values observed in the data

Value	Meaning
assistant	Assistant professor — the tenure-track rank that triggers all tenure-related outcomes
associate	Associate professor — typically indicates tenure has been granted
full	Full professor
endowed	Endowed / named chair
distinguished	Distinguished professor
lecturer	Non-tenure-track lecturer
senior_lecturer	Senior lecturer (non-tenure-track)
instructor	Instructor (non-tenure-track)
teaching_prof	Teaching-focused professor (non-tenure-track)
clinical	Clinical professor
adjunct	Adjunct faculty
postdoc	Postdoctoral researcher
fellow	Research or visiting fellow
visiting	Visiting faculty
research_prof	Research professor
research_scientist	Research scientist
senior_researcher	Senior researcher
scientist	Scientist
emeritus	Emeritus faculty
unknown	Rank could not be parsed from the page (most common reason: title ambiguous or missing)

Note for analysis: The tenure outcome model focuses on rows where `rank = assistant`. Rows with other ranks are retained in the panel for context (e.g. to confirm a promotion event) but are not the unit of analysis for the survival model.

Section 4 — Publications

Publication counts come from OpenAlex. They are **only populated when** `match_confidence` ∈ { `HIGH`, `MEDIUM`, `MULTI` }. Rows with `NONE` or `LOW` confidence have blank/zero publication counts and should be excluded from publication-based analyses.

Column	Type	Description
<code>pubs_year</code>	integer	Number of publications attributed to this person in OpenAlex for this specific calendar <code>year</code> .
<code>pubs_cumulative</code>	integer	Running cumulative publication count from the person's first observed year through the current <code>year</code> . Resets if the person's panel has gaps.

Section 5 — Tenure Track Spell

These columns describe the person's assistant-professor spell across all years in the data. They are the **same value repeated on every row** for a given person (person-level, not year-level).

Column	Type	Description
<code>ever_assistant</code>	boolean	<code>True</code> if the person was ever observed as an assistant professor in the data. <code>False</code> for people observed only at higher ranks (e.g. someone hired directly as an associate).
<code>first_asst_year</code>	integer	First calendar year the person appeared as <code>assistant</code> . Blank if <code>ever_assistant = False</code> .
<code>last_asst_year</code>	integer	Last calendar year the person appeared as <code>assistant</code> . Blank if <code>ever_assistant = False</code> .
<code>years_as_asst_so_far</code>	integer	In a given year-row, how many years the person has accumulated as assistant professor up to and including that year. Non-null only when <code>rank = assistant</code> in that row; blank for non-assistant-rank rows.

Section 6 — Tenure Outcomes

These are the three mutually exclusive outcome flags. Every person with `ever_assistant = True` falls into exactly one category: tenure, attrition, or censored.

Column	Type	Description
<code>tenure_event</code>	boolean	<code>True</code> if the person was observed transitioning to associate/full/endowed/distinguished rank within 2 years of their last assistant-professor year (<code>gap_tolerance = 2</code>). This is the primary outcome variable.
<code>year_of_tenure</code>	integer	Calendar year of the promotion event. Blank if <code>tenure_event = False</code> .
<code>attrition</code>	boolean	<code>True</code> if the person was last observed as an assistant professor and then disappeared from the data before the end of the observation window without a promotion. Interpreted as leaving academia or leaving the field.
<code>censored</code>	boolean	<code>True</code> if the person was still appearing as an assistant professor near the end of the observation window (within <code>gap_tolerance = 2</code> years of the last data year). Their outcome is unknown — they may yet be promoted or leave. Right-censored in survival analysis terms.

Survival analysis note: `tenure_event` is the “event,” `attrition` and `censored` are competing exit states. The Cox model treats censored observations appropriately; attrition is treated as a competing risk.

Section 7 — Data Quality

Column	Type	Description
<code>n_snapshots</code>	integer	Number of times the person's faculty page was successfully scraped in a given year. A value of 1 is typical; higher values occur when multiple scrape passes captured the page. Used internally to assess data reliability.
<code>match_confidence</code>	string	Quality tier of the OpenAlex author match. See table below.

Value	Meaning	Use in analysis
HIGH	Name match + institution found in OpenAlex affiliations + panel years overlap with affiliation years	Most reliable; use freely
MEDIUM	Name match + institution in affiliations, but year overlap was unclear	Reasonably reliable
MULTI	Multiple HIGH or MEDIUM candidates found — resolver picked the first one but the correct match is ambiguous	Use with caution; publication counts may belong to the wrong person
LOW	Name match only — institution was not found in OpenAlex affiliations	Weak match; treat publication counts as unreliable
NONE	No OpenAlex result returned for this person	No publication data; pubs_year and pubs_cumulative will be blank

Recommended filter for publication analyses: `match_confidence IN ('HIGH', 'MEDIUM')`. Excluding `MULTI` is conservative but appropriate for submitted work; including `MULTI` adds coverage but increases noise.

Section 8 — Peer Pool Metrics

The “peer pool” for person i in year t is defined as **all other assistant professors at the same institution in the same year** who also have OpenAlex publication data. These metrics capture the quality of the talent pool surrounding each individual — the core theoretical construct of the research.

LOO = Leave-One-Out. All pool statistics exclude person i themselves, so a person’s own publications do not inflate their own peer benchmark.

Column	Type	Description
<code>pool_size_all</code>	integer	Total number of assistant professors at the institution in that year (regardless of OpenAlex data availability).
<code>pool_size_oa</code>	integer	Number of assistant professors with OpenAlex data in that year (subset of <code>pool_size_all</code>).
<code>pool_size_oa_loo</code>	integer	Number of OpenAlex-matched assistants excluding person <i>i</i> . Equals <code>pool_size_oa - 1</code> if person <i>i</i> has OA data, otherwise equals <code>pool_size_oa</code> .
<code>poolq_loo_mean</code>	float	Leave-one-out mean of annual publications (<code>pubs_year</code>) among peer pool members. The primary “pool quality” regressor in the Cox model. Blank if <code>pool_size_oa_loo = 0</code> .
<code>poolq_loo_sd</code>	float	Leave-one-out standard deviation of annual publications in the peer pool. Blank if <code>pool_size_oa_loo < 2</code> .
<code>pool_rank_loo</code>	integer	Person <i>i</i> ’s ordinal rank within the full OA peer pool by publications (1 = lowest, <code>pool_size_oa</code> = highest). Blank if no OA data.
<code>pool_pctile_loo</code>	float	Person <i>i</i> ’s percentile (0–100) within the full OA peer pool. Blank if no OA data.

Quick-Reference: Recommended Filters for Common Analyses

Analysis goal	Suggested filter
Tenure outcome model (main analysis)	<code>ever_assistant = True</code>
Publication-based covariates	<code>match_confidence IN ('HIGH', 'MEDIUM')</code>
Conservative publication filter	<code>match_confidence = 'HIGH'</code>
Pool quality regressions	<code>pool_size_oa_loo >= 2</code> (need SD) or <code>>= 1</code> (need mean)
Exclude right-censored	<code>censored = False</code> (caution: biases survival estimates — use Cox model instead)
Assistant-professor rows only	<code>rank = 'assistant'</code>

